Rohan Hitchcock

rohan.hitchcock@pm.me • rohanhitchcock.com
linkedin.com/in/rohan-hitchcock • Google Scholar

Experience

Research Scientist 2026–Present
Resolution

Timaeus
Research Scientist 2026
Research Affiliate 2025–2026

Postgraduate Student 2023–2026
The University of Melbourne & CSIRO

Tutor (Teaching Assistant) 2020–2025
The University of Melbourne

The Peter Doherty Institute for Infection and Immunity
Research Assistant 2019–2020
Research Intern 2018–2019

Education

Doctor of Philosophy 2026
The University of Melbourne

Master of Science (with Distinction), Mathematics and Statistics 2022
The University of Melbourne

Bachelor of Science, Mathematics and Statistics 2020
The University of Melbourne

Diploma in Computing 2020
The University of Melbourne

Publications

"Singular Learning Theory for Deep Learning Interpretability: Stochastic Sampling Methods and Applications to Learning from Physical Systems" 2026
R. Hitchcock. *PhD Thesis, The University of Melbourne.*

"Susceptibilities for Neural Networks Learning from Physical Data" 2026
R. Hitchcock, G. W. Delaney, J. H. Manton, R. Scalzo and J. Zhu. *Advances in Neural Information Processing Systems (NeurIPS 2026).*

"From Global to Local: A Scalable Benchmark for Local Posterior Sampling" 2025
R. Hitchcock and J. Hoogland. *arXiv:2507.21449.*

"Emergence of Computational Structure in a Neural Network Physics Simulator" 2025
R. Hitchcock, G. W. Delaney, J. H. Manton, R. Scalzo and J. Zhu. *XAI4Science Workshop at the International Conference on Learning Representations (ICLR 2025).*

"Differentiation, Division and the Bicategory of Landau–Ginzburg Models" 2022
R. Hitchcock. *MSc Thesis, The University of Melbourne.*

"Display of Native Antigen on cDC1 That Have Spatial Access to Both T and B Cells Underlies Efficient Humoral Vaccination" 2020
Y. Kato, T. M. Steiner, H.-Y. Park, **R. Hitchcock** et al. *Journal of Immunology* 205(7):1842–1856.

Research Notes

"Interpretability as a Benchmark for Sampling" 2026
A. Gordon, **R. Hitchcock**, I. Mouratidis, J. Wind and D. Murfet. *Resolution.*

"Guide for Sampling Hyperparameter Selection" 2026
R. Hitchcock, J. Hoogland, G. Wang and A. Gordon. *Timaeus.*

"Interpreting the Ising Model" 2026
A. Gordon, **R. Hitchcock** and D. Murfet. *Timaeus.*

"Spectroscopy at Scale: Finding Interpretable Structure in Pythia-1.4B" 2026
D. Murfet, A. Gordon, M. Adam, G. Wang, J. Hoogland, G. Baker, W. Snell, S. van Wingerden, A. Newgas, B. Snikkers and **R. Hitchcock**. *Timaeus.*

Talks

"Sampling Hyperparameter Selection in SLT" 2026
ILIAD, Berkeley

"Towards Higher Confidence in Alignment" 2026
Australian AI Safety Forum, Sydney

"Sampling from Degenerate Loss Landscapes" 2025
Focus Period: Mathematical Science of AI Safety, Sydney Mathematical Research Institute

"The Empirical Process in Singular Learning Theory" 2025
SLT Seminar, online

"Local Learning Coefficient Estimation: The Story So Far and Open Problems" 2025
ILIAD, Berkeley

"Induction Heads and Phase Transitions" 2023
SLT & Alignment Summit, Berkeley